

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I VAIBHAVAK LAKSHMI K (192511310) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

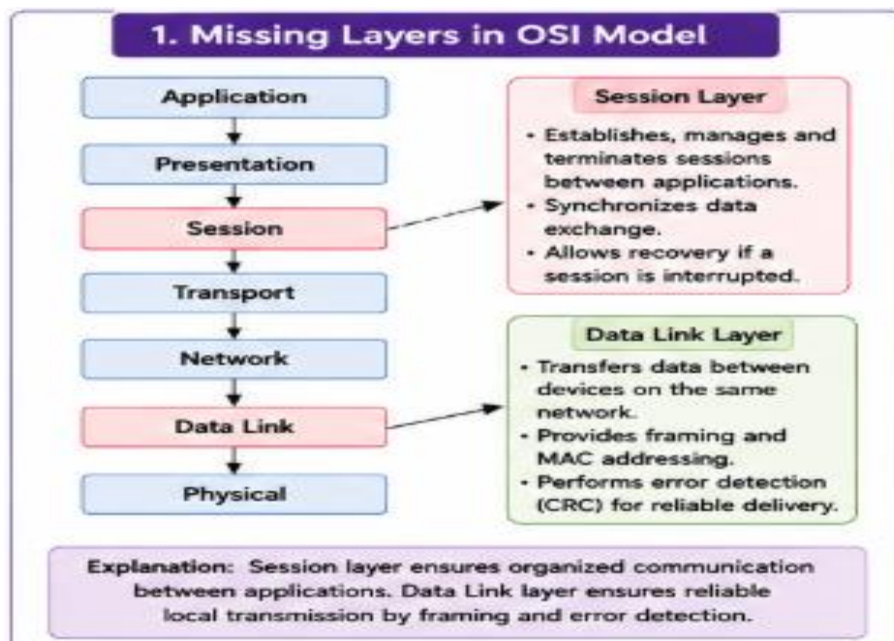
CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

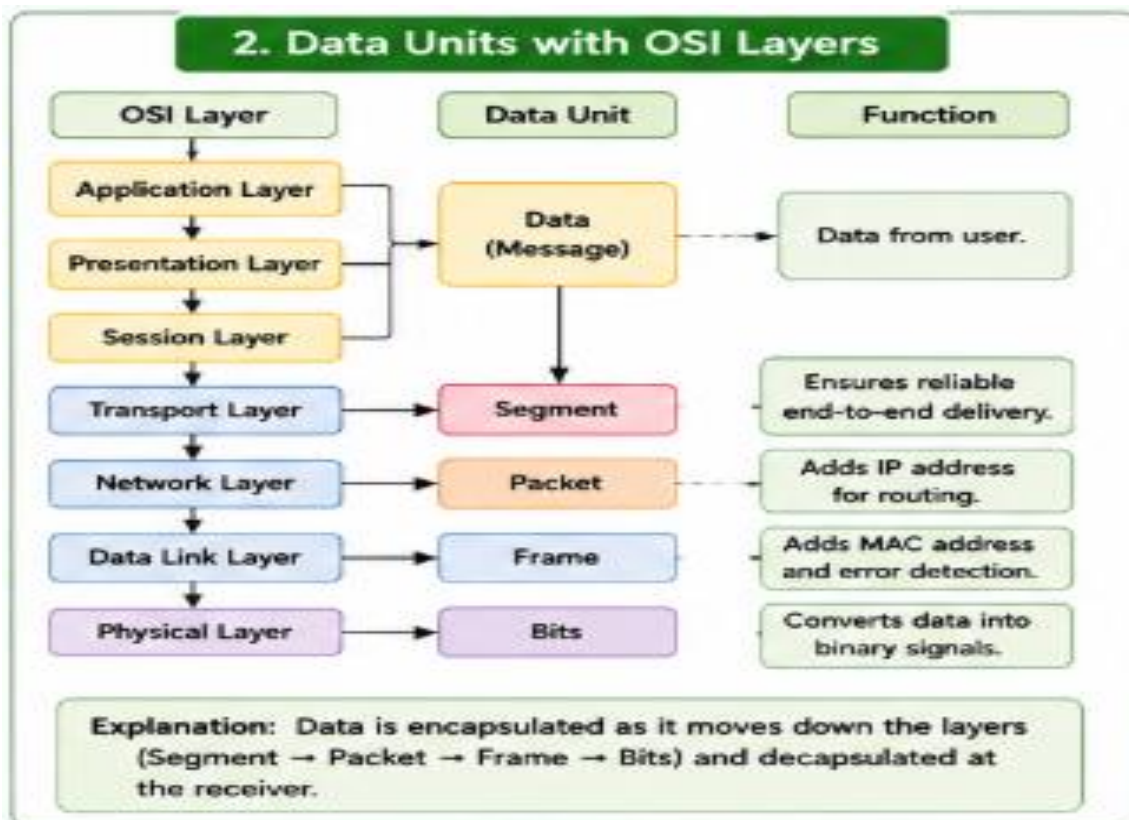
Answer:



These layers are essential because the Session layer ensures smooth communication between applications, while the Data Link layer provides reliable local transmission by detecting transmission errors. Together, they improve the reliability of data communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Answer:



This process is called **encapsulation**. At the receiving end, the reverse process called **decapsulation** converts Bits → Frame → Packet → Segment → Data. This transformation ensures proper routing, error detection, and successful delivery of information.

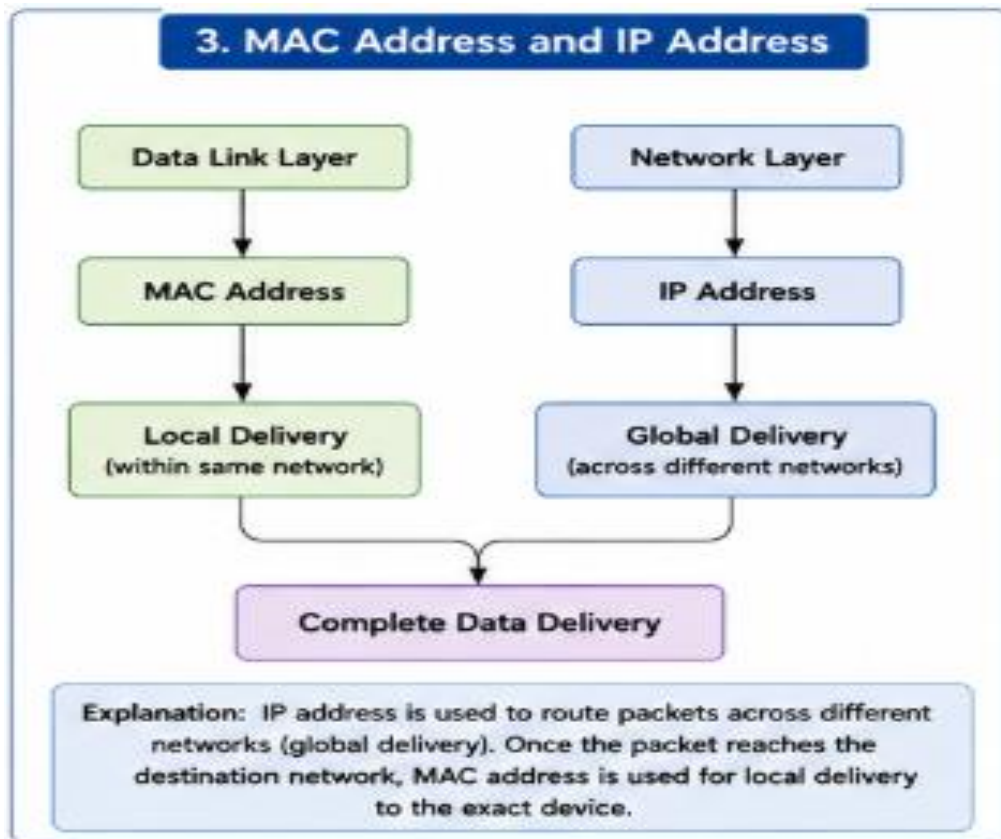
3. Given the concept map:

DataLink→MACAddress→LocalDelivery Network

→ IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Answer:



The **Data Link Layer** uses **MAC addresses** to identify devices within the same Local Area Network (LAN). MAC addresses provide local delivery between directly connected devices.

The **Network Layer** uses **IP addresses** to identify devices across different networks. Routers use IP addresses to forward packets to the correct destination network.

When a packet reaches the destination network, the MAC address is used again to deliver the frame to the exact destination device.

Therefore, the combination of IP addressing for global routing and MAC addressing for local delivery ensures complete and accurate communication across networks.

4. A concept map shows:

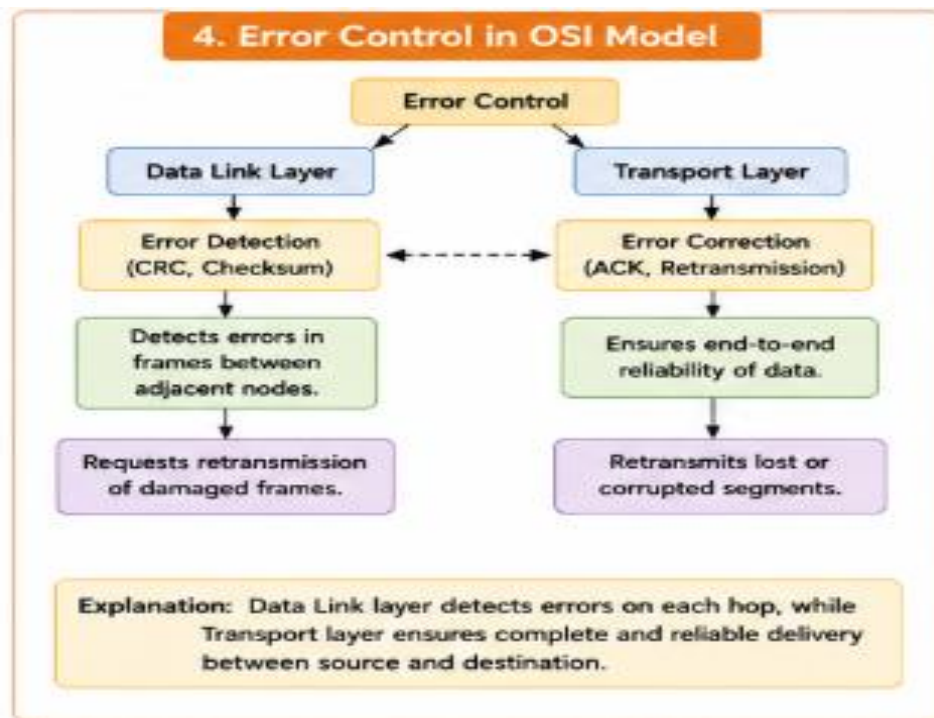
ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Answer:



Explanation:

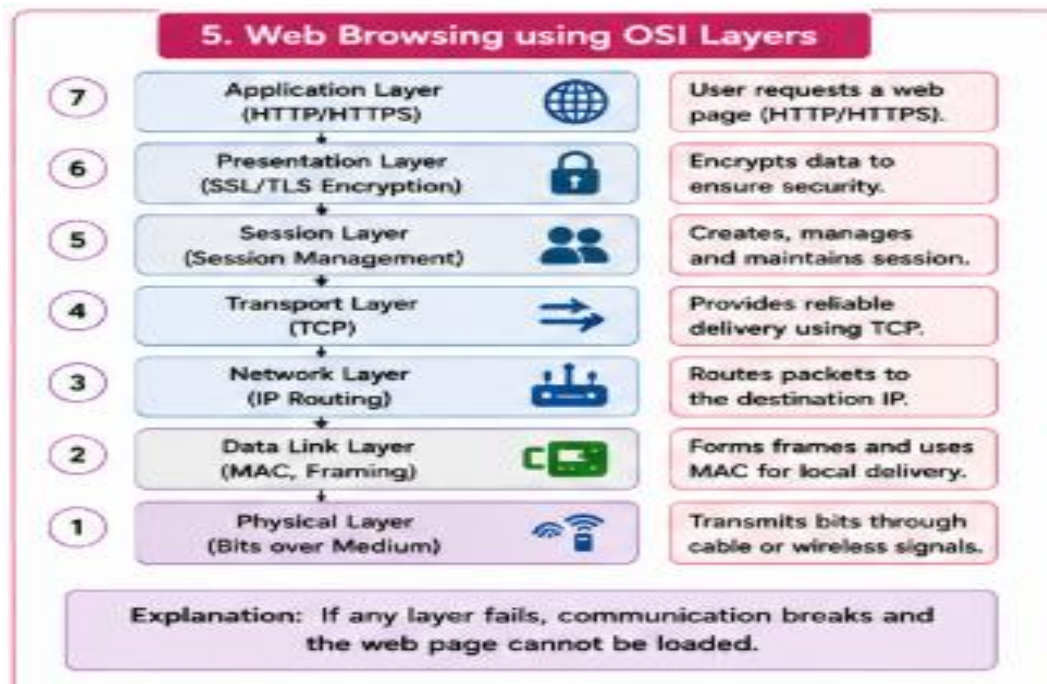
The **Data Link Layer** detects transmission errors using techniques such as CRC (Cyclic Redundancy Check). If an error is detected, damaged frames can be retransmitted between neighboring devices.

The **Transport Layer** provides end-to-end reliability by using acknowledgements (ACKs), sequence numbers, and retransmission of lost segments.

The Data Link Layer protects communication between adjacent devices, while the Transport Layer guarantees reliable communication between the sender and receiver. Together, these layers reduce data loss, improve accuracy, and ensure reliable communication.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Answer:



Explanation:

When a user opens a website, the Application Layer sends an HTTP request. The Presentation Layer encrypts the data using SSL/TLS. The Session Layer maintains the communication session. The Transport Layer ensures reliable delivery using TCP. The Network Layer routes packets using IP addresses. The Data Link Layer frames the data and uses MAC addresses for local delivery. Finally, the Physical Layer transmits the data as bits.

If any layer fails, communication is interrupted.

- Failure of the Physical Layer disconnects the network.
- Failure of the Data Link Layer prevents local communication.
- Failure of the Network Layer stops packet routing.
- Failure of the Transport Layer causes incomplete or unreliable data transfer.
- Failure of the Session Layer disconnects communication.
- Failure of the Presentation Layer affects encryption and data translation.
- Failure of the Application Layer prevents the web page from loading.

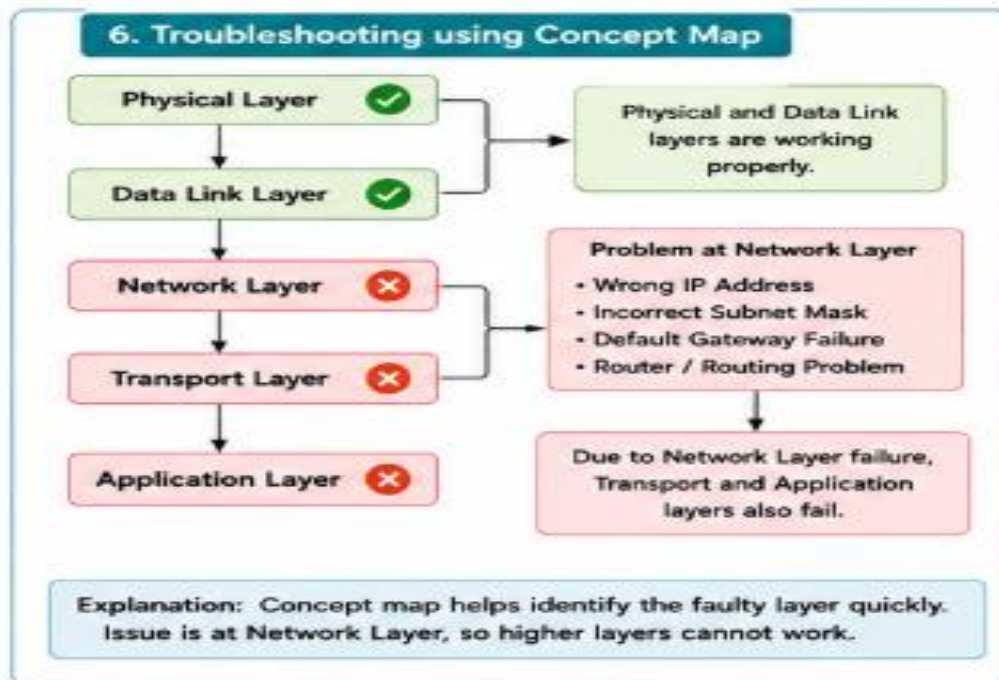
Thus, every OSI layer depends on the services of the lower layer, making all layers essential for successful web browsing.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

Answer:



RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation